

VENTUNO Q is the definitive powerhouse for AI and robotics applications.



ARDUINO[®] VENTUNO[™] Q

Where AI takes action

The edge AI computer that redefines what's possible.

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ARDUINO

VENTUNO Q

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With VENTUNO Q, AI gets physical

Optimized for local LLM, agentic AI, and computer vision, empowering AI users to become AI creators.

Dual brain

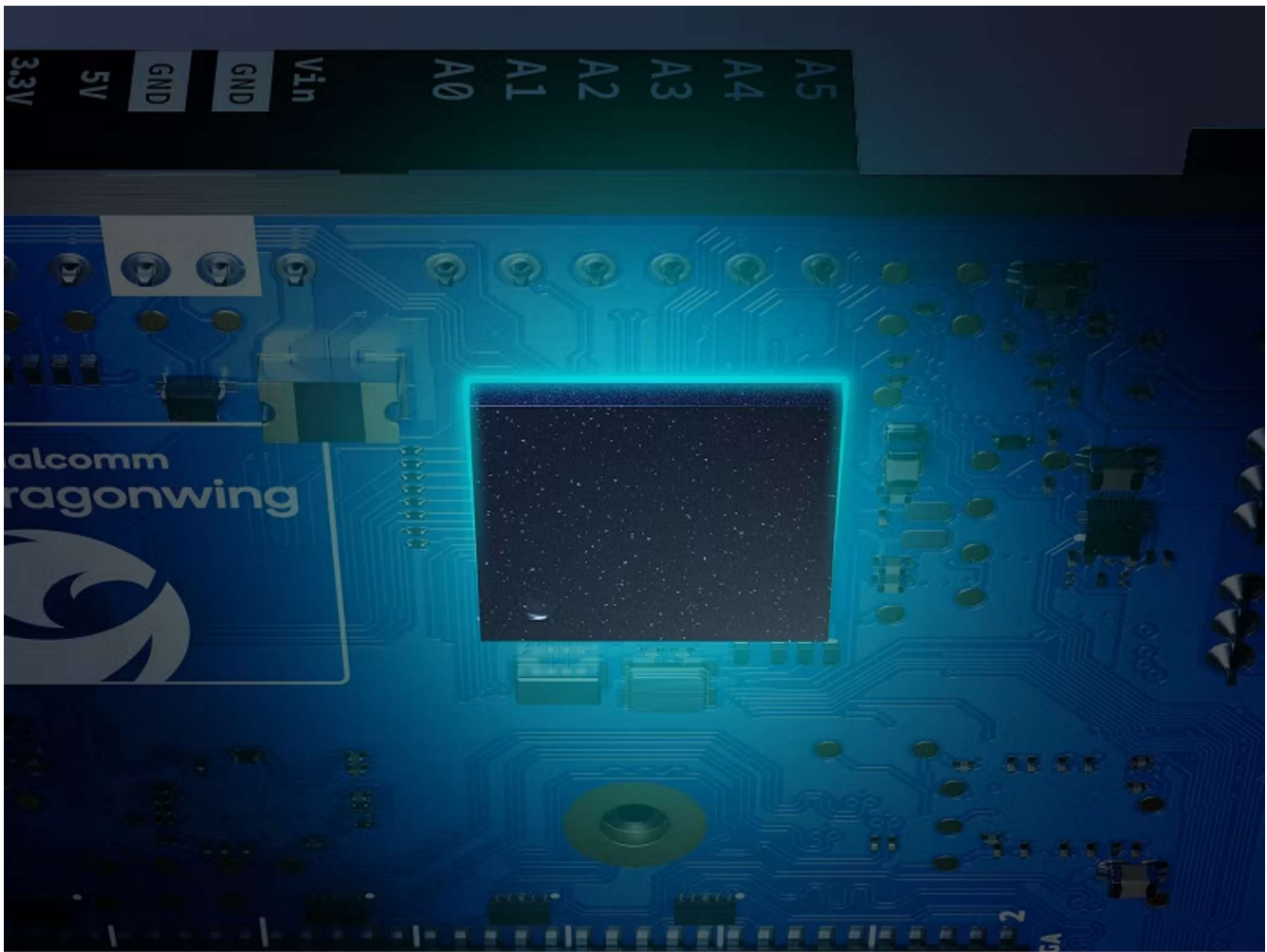
The accelerated dual-brain architecture for real-world AI

VENTUNO Q brings perception, decision, and action onto a single board – eliminating complexity, latency, and the cost of multi-device setups. It combines a Qualcomm Dragonwing[™] IQ8 processor and STM32H5 microcontroller, which communicate seamlessly via RPC (Remote Procedure Call) bridge.



The AI Brain

The Dragonwing IQ-8275 processor delivers NPU, CPU, and GPU for complex neural network inference – with an outstanding 40 dense TOPS.



The Action Brain

The STM32H5F5 microcontroller enables sub-millisecond response to guarantee stable, deterministic control for robotics, motion systems, and industrial interfaces.

ENGINE SPECIFICATIONS

Uncompromising silicon & software

An integrated ecosystem mapping edge intelligence directly onto robust, physical actuation units.

AI engine

40

40 dense Tops

Qualcomm® HEXAGON™

Memory

16

16 GB LPDDR5 RAM

Shared by NPU, CPU, and GPU

CPU

x8

Octa-core

Qualcomm® Kryo™ up to 2.36 GHz
Real-Time

<1

<1 ms

STM32H5F5 at 250 MHz
ProcessorDragonwing IQ8
MicrocontrollerSTM32H5
GraphicsQualcomm® Adreno™ 623
ExpansionM.2 Slot for NVME (PCIe Gen.4)
OSUbuntu, Zephyr (and Debian coming soon)

UNIFIED DEV EXPERIENCE

Arduino® App Lab: one platform to play, train, and master

Develop Arduino sketches, Python scripts, and AI models in one consistent environment.

Arduino App Lab seamlessly bridges embedded programming, Linux development, and edge AI, giving you a single environment for your entire application stack.

[GET ARDUINO APP LAB](#)

Versatile, ready-to-run AI models open infinite possibilities

Accelerate development with a wide range of AI models optimized for the integrated NPU of VENTUNO Q – or customize for your unique challenges! Powered by Edge Impulse and Qualcomm® AI Hub, this is your gateway to extensive AI libraries.

Why do leaves change color in the fall? ...

Leaves change color in the fall because of changes in daylight and temperature. As days get shorter and cooler, trees begin to slow down their food production.

The green pigment, chlorophyll, breaks down, revealing other pigments like carotenoids and anthocyanins. These pigments give leaves their

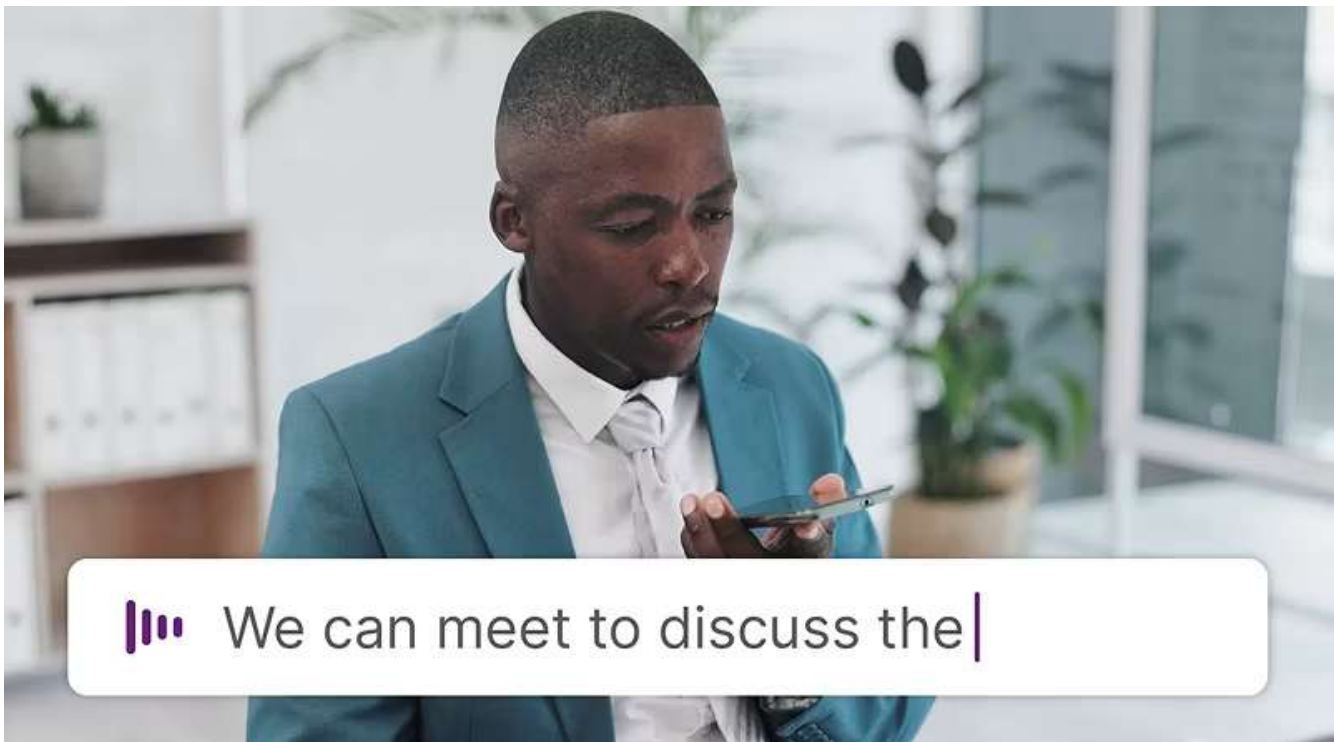
Local LLMs

Use **Qwen 4** or **Gemma 4** for sophisticated natural language understanding entirely on-device, without cloud dependencies or data transmission.



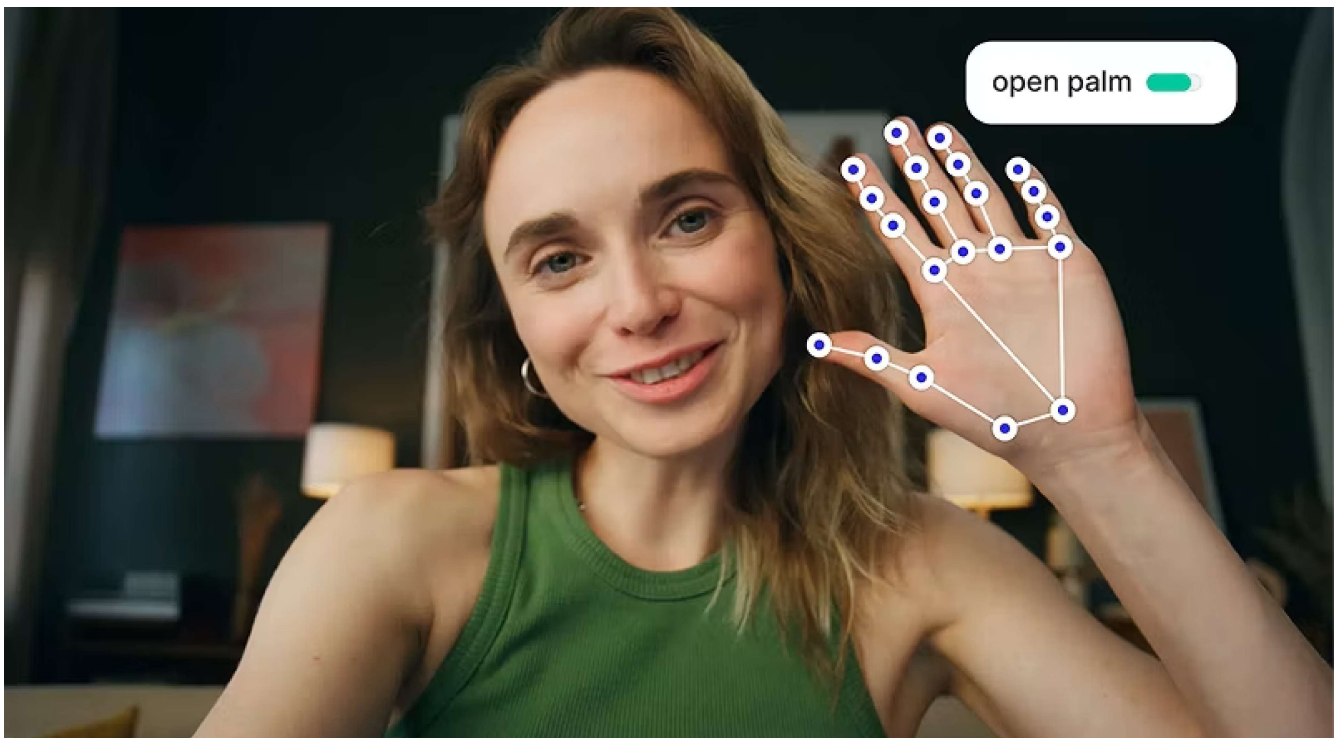
Local VLMs

Models like **Qwen 3 VLM** combines visual perception with natural language understanding for image captioning, scene description, OCR, and more.



TTS & ASR

Leverage **Melo TTS** and **Whisper** to allow offline devices to understand natural speech, transcribe, and give human-like responses.



Gesture recognition

Recognize hand gestures, finger movements, and sign language for touchless interfaces and human-robot interaction using **MediaPipe**.



Object tracking

Use **YOLO-X** (You Only Look Once - eXtended) to track people, vehicles, or objects in real-time across multiple camera views.



Pose detection

Track body pose, joint positions, and movement patterns with models like **PoseNet**. For fitness applications, safety monitoring, and interactive gaming.

4 ways you can build with AI



Play

Open a curated example and watch AI run in real time, no code required.



Tinker

Modify a ready-to-use example: adjust parameters, tweak the code, instantly update your app.



Shape

Bring your own model from Hugging Face or Qualcomm AI Hub, or train one with Edge Impulse.



Master

Customize using your own toolchains, and build advanced pipelines with the full power of Linux.

Stay tuned for more updates in AI Models

[PRE-ORDER NOW](#)

ROBOTICS-READY

Purpose-built for robotics and physical actuation

VENTUNO Q is engineered specifically for systems that move, manipulate, and respond to the physical world with precision and reliability.

- **ROS 2 compatibility**

VENTUNO Q supports ROS 2 (Robot Operating System 2) for advanced, real-time robot development.

- **Robotics Bricks in Arduino App Lab**

Arduino App Lab includes specific robotics Bricks that bundle complex functionality into accessible, reusable components.

- **Real-time motion control excellence**

Instant response on GPIO, PWM, and CAN-FD enables zero-jitter motor control and safety-critical reliability.

Build your own

Real-world Applications



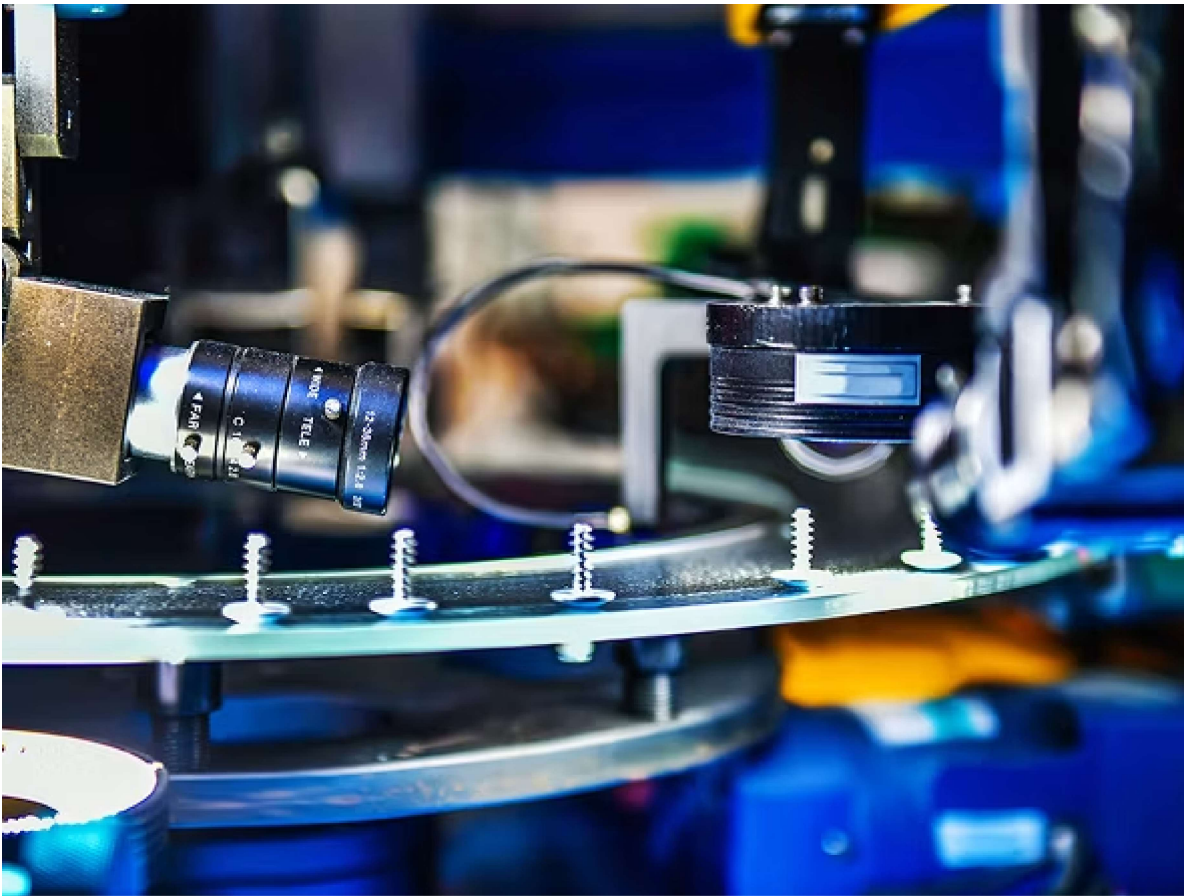
AI-powered systems

AI-capable assistants that operate entirely offline, ensuring complete data privacy. Perfect for smart kiosks, healthcare assistants, traffic flow analysis, and more.



Robotics and motion control

The complete robotics stack: vision processing for environmental awareness combined with deterministic motor control for precise manipulation and navigation.



Edge AI vision and sensing systems

Automated quality control, predictive maintenance, and process optimization: all while meeting industrial reliability, safety, and connectivity standards.

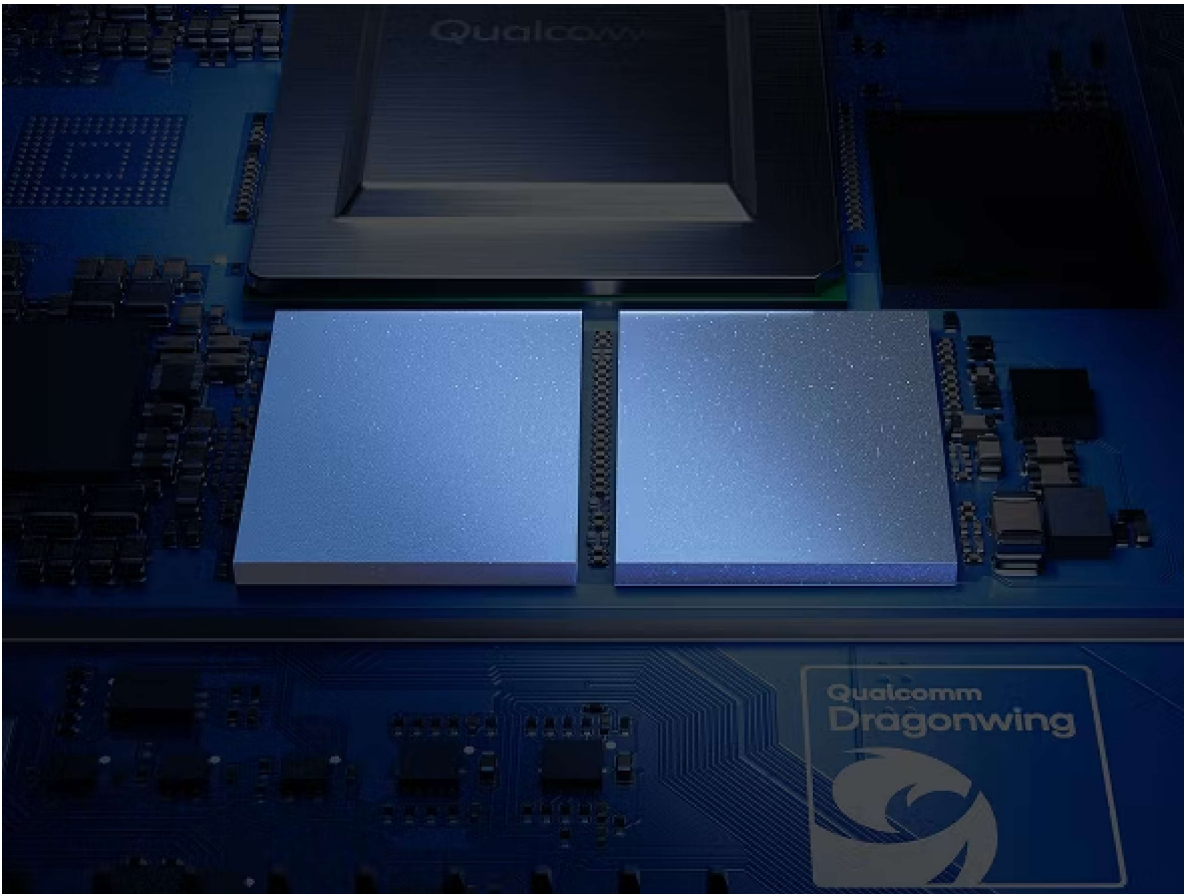


Education & research

A powerful, flexible platform for prototyping algorithms, publishing breakthrough research, and teaching advanced AI and robotics concepts.

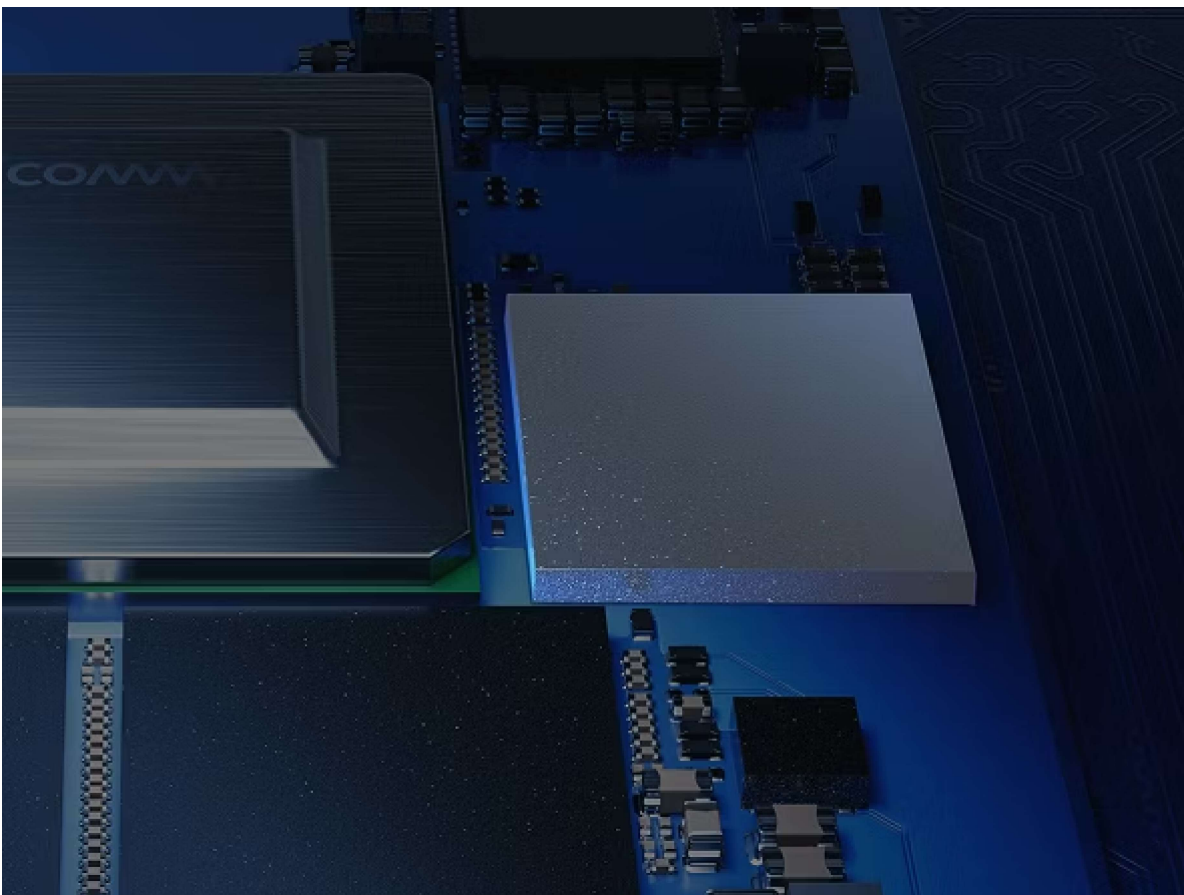
Explore all the spectacular specs

VENTUNO Q was designed with all the hardware features to make it a powerful and accessible edge AI platform, ready to go from prototyping to production with zero friction.



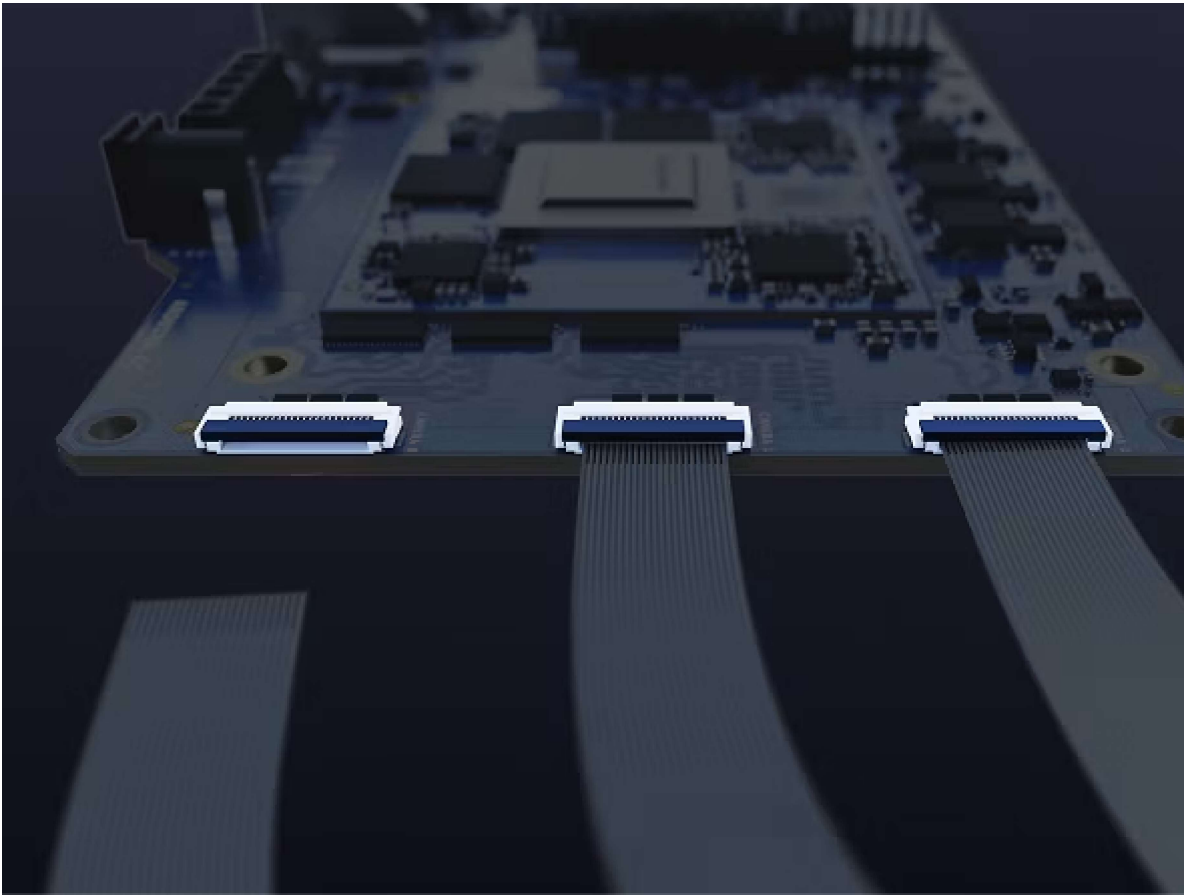
16 GB RAM

High-bandwidth 16 GB RAM enables concurrent multi-model AI inference.



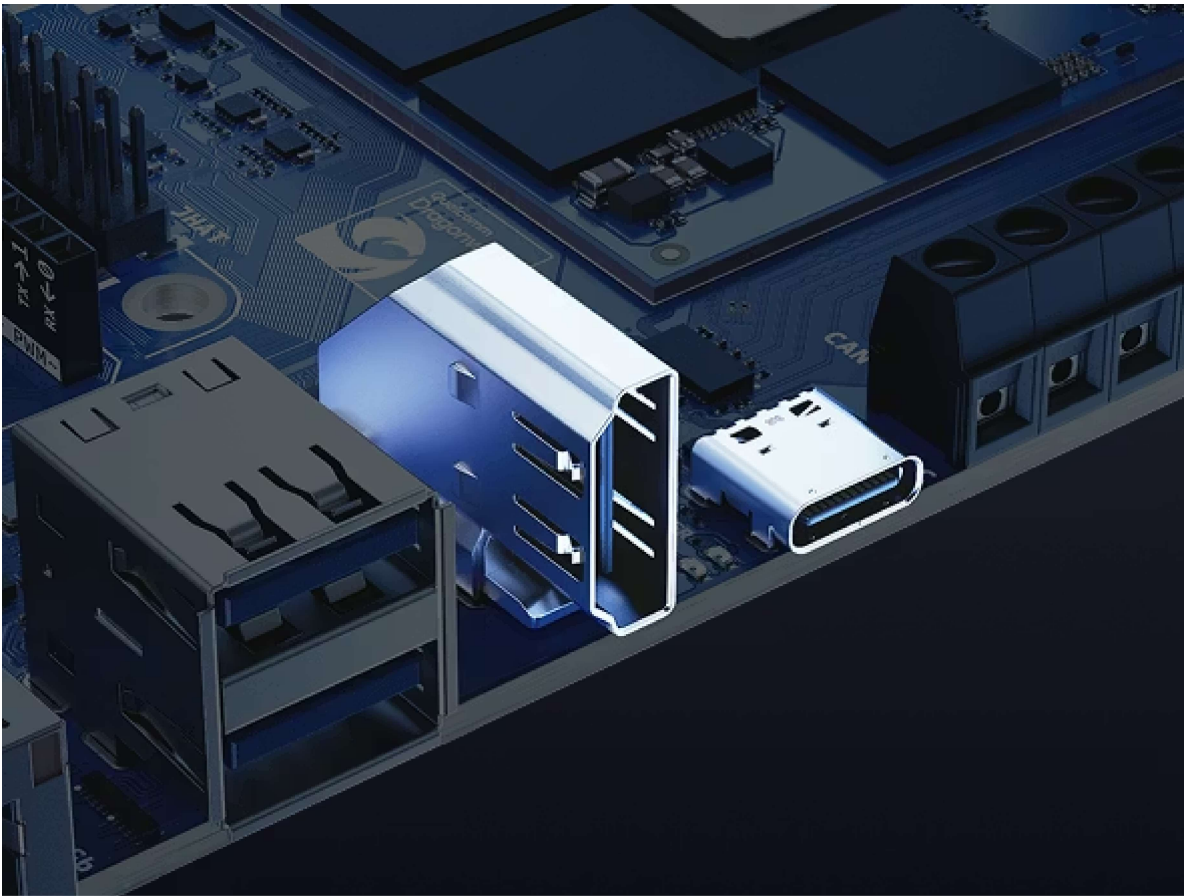
64 GB expandable storage

Industrial-grade eMMC, plus M.2 connector for NVMe Gen.4 storage expansion.



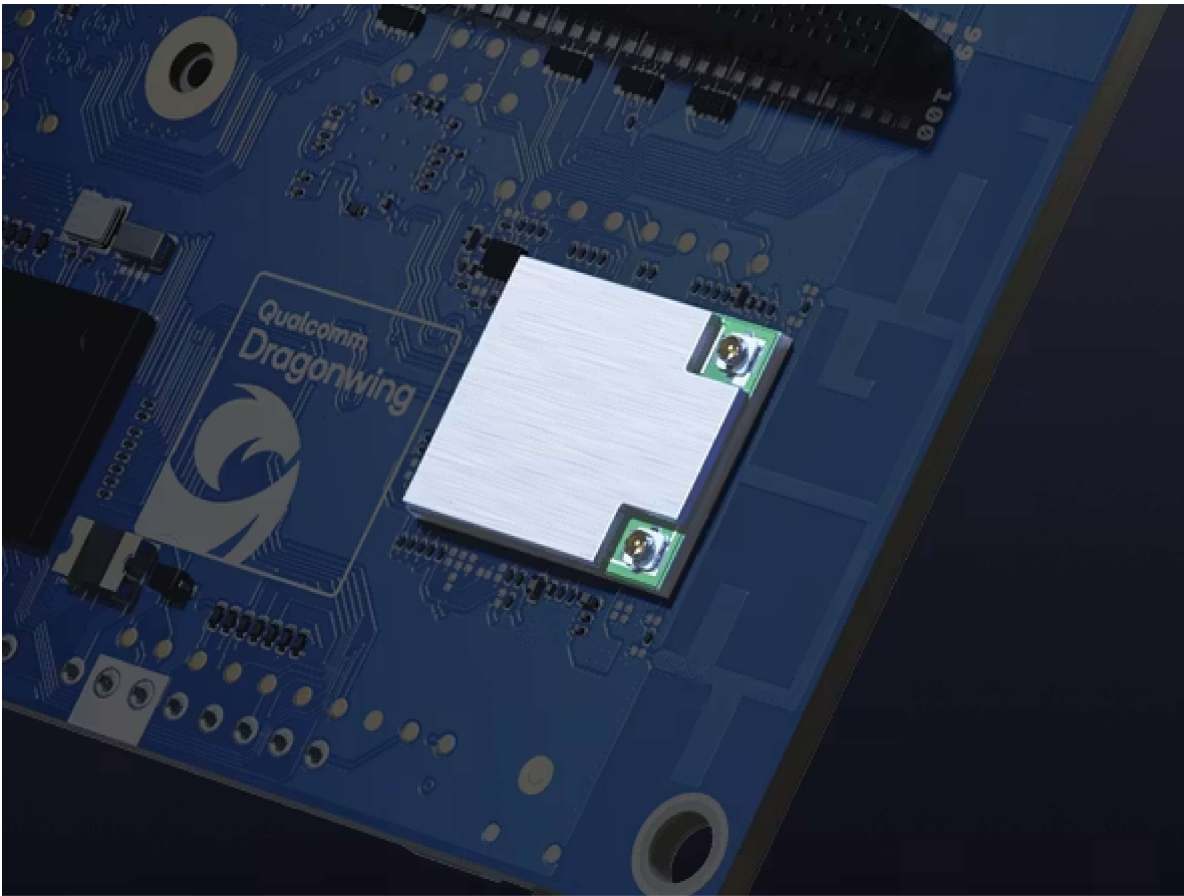
Triple MIPI-CSI camera support

Three 4-lane interfaces for 360° awareness, stereo depth perception, and multi-angle inspection.



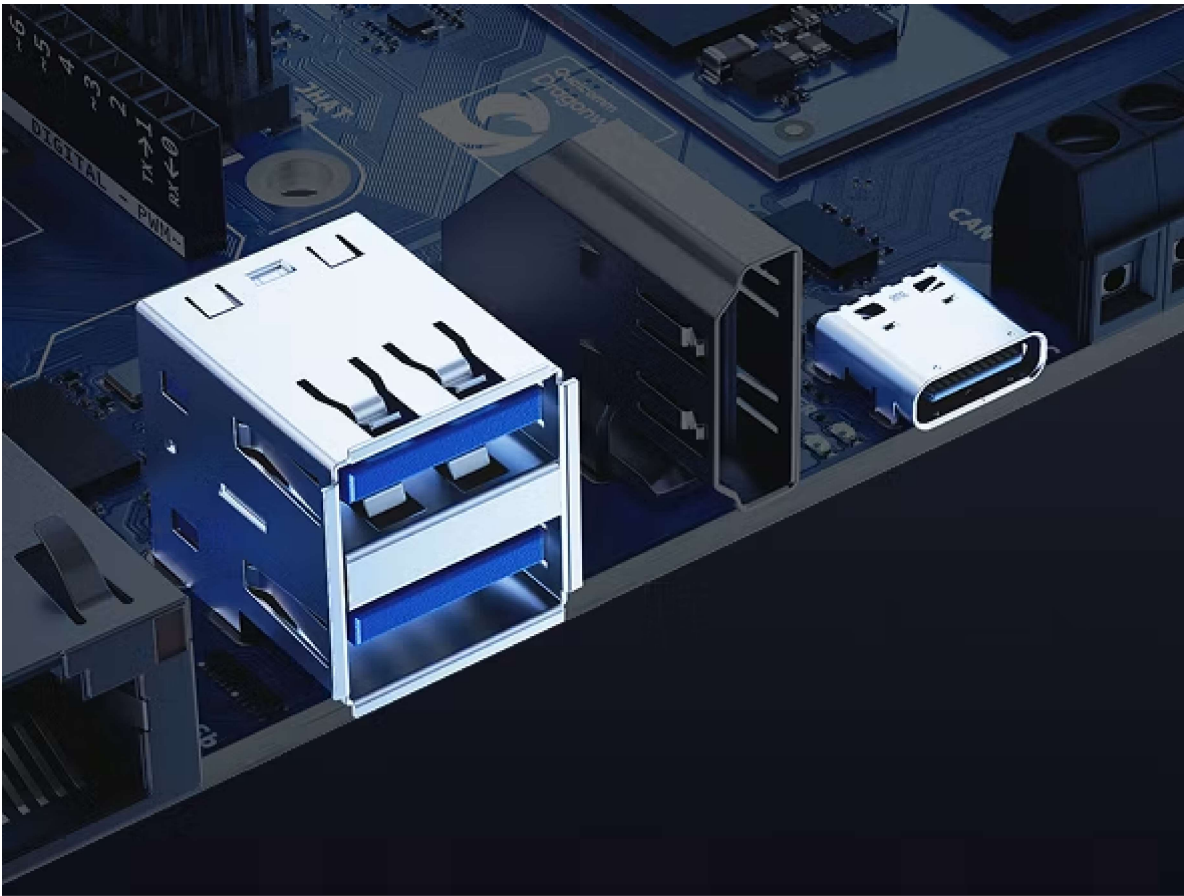
Display outputs

MIPI-DSI for touchscreens and interactive panels, HDMI for monitors and projectors, USB-C DisplayPort Alt Mode for versatile video output.



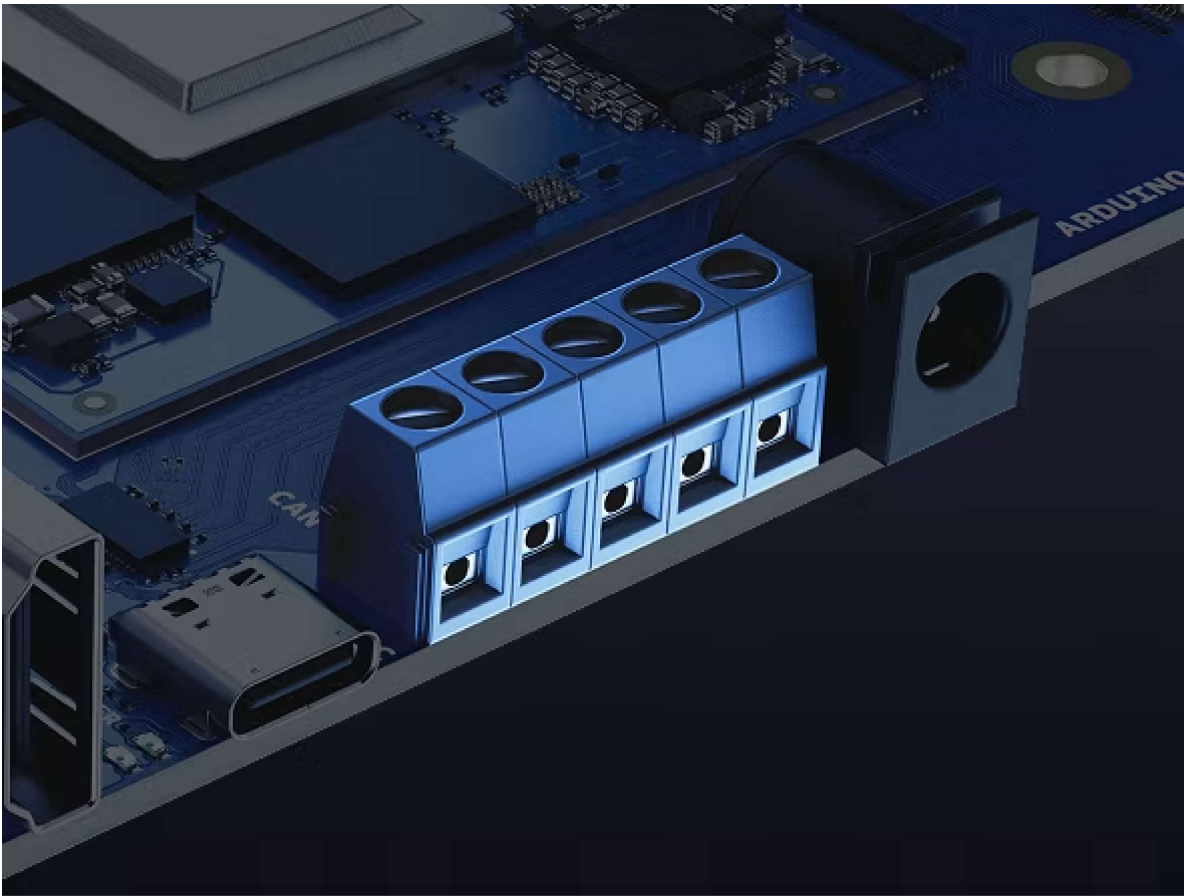
Wireless & wired networking

Tri-band Wi-Fi® 6 (2.4/5/6 GHz) and Bluetooth® 5.3 for wireless connectivity. 2.5 Gb Ethernet.



Universal USB connectivity

USB-C for high-speed vision and audio data transfer. Dual USB-A 3.0 ports for peripherals and storage.



Industrial I/O & control

CAN-FD for professional motor controllers and industrial networks. PWM and deterministic GPIO for sub-millisecond response.

Compatibility

Unmatched hardware compatibility for rapid innovation

VENTUNO Q delivers unmatched ecosystem compatibility, extensive memory and storage, and professional-grade expansion options. Because Innovation accelerates when you have the right tools at your fingertips.

 Arduino® UNO™ shields support

Arduino® UNO™ shields support

Native compatibility with all UNO shields, motor controllers, sensors, and wireless interfaces work out of the box.

 Raspberry Pi® Hat compatibility

Raspberry Pi® Hat compatibility

Standard 40-pin GPIO header provides full electrical and mechanical compatibility with Raspberry Pi accessories.

 Arduino® Modulino™ nodes & Qwiic connectivity

Arduino® Modulino™ nodes & Qwiic connectivity

Solder-free Qwiic connector supports chainable Modulino nodes and hundreds of Qwiic-compatible sensors with polarized, error-proof connections.

 High Speed Media Carrier Support

High Speed Media Carrier Support

Advanced carrier headers (JMEDIA, JOMEGA, JMISC) provide dedicated lanes for high-speed cameras, displays, multi-axis control, and sensor fusion.

Works with Arduino

Prototype on Arduino, scale to production

Works with Arduino™ certifies third-party System-on-Modules as fully compatible with the Arduino ecosystem. Any app you build in Arduino App Lab on UNO Q or VENTUNO Q runs on a certified SOM with zero code changes. A guaranteed production path for developers, and a credible badge for SOM vendors.

[Discover More](#)

Componente	Specifiche
Microprocessor (MPU)	Qualcomm Dragonwing™ IQ8 (IQ-8275): • CPU: 8-core Qualcomm® Kryo™ • GPU: Qualcomm® Adreno™ 623 • NPU: Qualcomm® Hexagon™ 40 dense TOPS • Qualcomm Spectra 692 ISP OS: Ubuntu or Debian upstream
Microcontroller (MCU)	STM32H5F5: • Arm® Cortex® M33 at 250MHz • 4MB flash



Frequently Asked Question

Where can I buy it?

VENTUNO Q is available from the [Arduino Store](#) and official distribution partners – [DigiKey](#), [Farnell](#), [Mouser](#), [Robu.in](#), [RS](#), and many others worldwide.

Can I develop with standard Linux tools, instead of the Arduino App Lab?

Yes, VENTUNO Q is a real, standard Ubuntu computer. This means you can use standard IDEs such as VS Code, PyCharm, Eclipse, Vim, Emacs, Python virtual environments and package managers, Docker containers for isolated environments, SSH remote development and headless operation. Any framework or toolchain you prefer! Arduino App Lab is the fast path, not a requirement.

Which AI models does VENTUNO Q support?

VENTUNO Q delivers up to 40 dense TOPS of AI compute, supporting Qwen 3 4B LLM; Qwen 2.5 7B and 3 4B VLM; Gemma 4 E2B and E4B; Whisper ASR; Melo and Piper TTS; YoloX small object detection; MediaPipe gesture recognition; and more.

What frameworks does VENTUNO Q support?

Get started with ready-to-run Bricks, optimized for the onboard NPU through Qualcomm® AI Hub. For generative AI use cases, VENTUNO Q supports LLM inference via llama.cpp and GGUF models, ideal for running locally at the edge. You can also run these models with GenieX, the open-source Gen AI runtime by Qualcomm. At the same time, you can work with familiar frameworks like PyTorch and other advanced AI stacks, enabling custom model development, optimization, and deployment tailored to specific applications.

What makes VENTUNO Q different from other edge AI boards?

VENTUNO Q is designed to take you from idea to deployed application – not just working prototype. At the heart of this experience is Arduino App Lab, a complete end-to-end platform for building, deploying, and managing AI applications that goes well beyond what any other edge AI board offers today, making professional-grade edge AI accessible at every skill level.

A dual-brain architecture of VENTUNO Q combines up to 40 dense TOPS of AI compute with a dedicated real-time microcontroller, enabling both intelligent inference and deterministic actuation in a single board. This is ideal for robotics, industrial control, and any application where AI must interact with the physical world. Broad ecosystem compatibility with UNO shields, Raspberry Pi Hats, and Modulino nodes keeps prototyping fast and familiar.

Which operating systems does VENTUNO Q support?

VENTUNO Q ships with pre-loaded Ubuntu, distributed by Canonical, and bundled with Ubuntu Pro license that delivers enterprise-grade security updates - which make it immediately viable for professional, commercial, and industrial developers. It supports **Debian** (coming soon) with full **upstream**. It supports for embedded-focused development. The microcontroller side of its dual architecture runs the Arduino Core on **Zephyr RTOS**.

How much RAM and storage does VENTUNO Q have?

VENTUNO Q comes with 16 (2 X 8) GB of LPDDR5 RAM and 64 GB of eMMC onboard storage, working together so you can run real AI on the board — on the edge and offline.

16 GB RAM is your fast working memory — enough to load generative models (LLMs and VLMs like Qwen 3 and Gemma 4, plus larger GGUF models), run inference, and keep your app and Ubuntu going all at once. That's roughly double the memory of typical 8 GB edge boards, so you can work with bigger models and richer multimodal pipelines without hitting a ceiling.

64 GB onboard storage is where Ubuntu, your AI models, your application, and your data live. Because VENTUNO Q boots and runs from fast, soldered-down storage rather than a removable microSD card, you get reliable, repeatable boots and performance built to ship and stay in the field.

How much storage can I add beyond the onboard 64 GB?

The M.2 connector supports NVMe SSDs.

Can I train my own custom AI models for VENTUNO Q?

Absolutely. You can train your own models on 3rd-party and custom inference engines. We recommend using Edge Impulse's integrated platform to upload your training data, train

models on the cloud infrastructure, and automatically optimize and quantize for VENTUNO Q hardware – then you can deploy directly into Arduino App Lab with one click!

Does VENTUNO Q support ROS (Robot Operating System)?

Yes, VENTUNO Q is compatible with ROS 2.

Can I take a VENTUNO Q prototype to production?

Yes. Through the [Works with Arduino™](#) Program, VENTUNO Q gives you a clear path from prototyping to production. Develop and validate on VENTUNO Q, then move to a third-party SOM built on the same Qualcomm Dragonwing™ IQ8 — scaling your product while preserving your software investment, since compatible Arduino Apps port across with zero code changes. Certified partners include SECO, with its [SOM-SMARC-Dragonwing-IQ8](#) and Clea software platform, and Toradex, with its [Aquila IQ8](#) and Torizon platform. And more OEMs are collaborating with Arduino and Qualcomm to bring additional Works with Arduino certified products to market.

Learn more about the [Works with Arduino Certification Program](#).

 Join a community of 33M+

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The future of edge AI and robotics isn't just intelligent, it's actionable. VENTUNO Q delivers the compute power, real-time control, and development ecosystem you need to transform your ideas into production-ready systems.

 Join a community of 33M+

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